AI Brain Tumor Analysis Report

File: test_pituitary.jpg

Sequence: T1-Weighted Contrast Enhanced

Patient: Patient A ID: 001 | Generated: 2026-03-26 10:13:25

SEVERITY: MILD | PREDICTION: PITUITARY | CONFIDENCE: 73.68% | GROWTH RISK: LOW

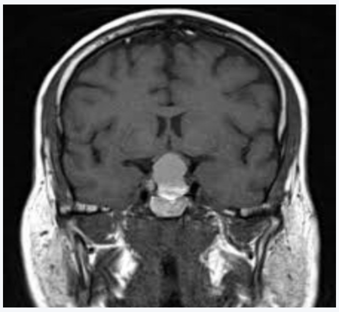
ANALYSIS SUMMARY

Tumor Type:	PITUITARY	**Tumor Area:**	8.525 cm2	**Laterality:**	Bilateral/Midline
Confidence:	73.68%	**Diameter:**	3.295 cm	**Midline Shift:**	0.1 mm
Uncertainty:	+/-14.2%	**Est. Volume:**	4.263 cm3	**Compression:**	None
Calibration:	0.858	**Brain Coverage:**	3.85%	**Sulcal Eff.:**	Absent
Severity:	Mild	**Bounding Box:**	2.85 x 4.15 cm	**Aggressiveness:**	0.300 / 1.0

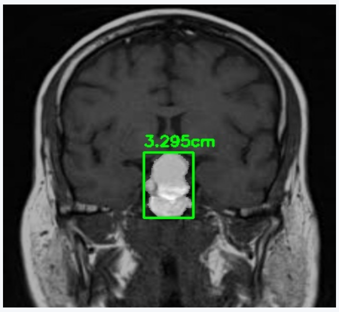
LESION-AWARE MULTI-MODEL FUSION

Decision:	Accept fused output	**EfficientNetV2-S:**	Pituitary	**MobileNetV3:**	Pituitary	**ConvNeXt Tiny:**	Pituitary
Agreement:	100%	**Scan Quality:**	0.9244	**Reliability:**	0.7236	**XAI Consistency:**	0.2708

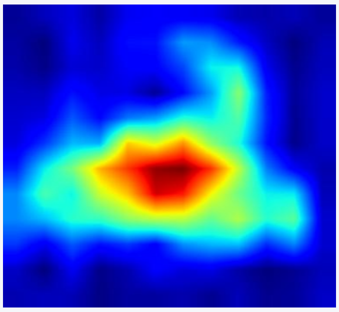
VISUAL ANALYSIS



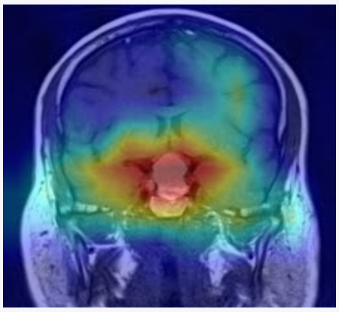
Original MRI



Segmentation



GradCAM Heatmap



GradCAM Overlay

SHAPE ANALYSIS | MASS EFFECT | RISK SCORE

Irregularity:	0.423	**Laterality:**	Bilateral/Midline	**Aggress. Score:**	**0.300 / 1.0**
Compactness:	1.734	**Shift:**	0.1 mm	**Growth Risk:**	**Low**
Convexity:	0.902	**Compression:**	None	**Progression:**	17%
Border:	Moderately defined	**Sulcal:**	Absent	**Uncertainty:**	+/-14.2%
Roughness:	Moderate				

PRIOR-CASE COMPARISON

Progression Flag:	**Progression**	**Area Change:**	**+700.5%**

RANO CRITERIA ASSESSMENT

Size Category:	Large	**Necrosis:**	Not clearly identified
Enhancement:	Heterogeneous	**Grade:**	**High-Grade (imaging features suggest possible high-grade - histopathological confirmation required)**

CLINICAL DECISION SUPPORT

Urgency Level: **HIGH**

- >> [DR-GATE] Specialist review required before relying on automated result
- >> PET scan consideration
- >> Neurosurgical consultation
- >> Neurological evaluation

Quantitative breakdown of tumor characteristics derived from segmentation and classification models.

KEY METRICS AT A GLANCE

Tumor Area	Diameter	Volume	Brain Cov.	Midline Shift	Risk Score
8.525 cm2	3.295 cm	4.263 cm3	3.85%	0.1 mm	0.300 / 1.0

SHAPE DETAIL

Irregularity: 0.423
Higher = more irregular boundary

Compactness: 0.495
Normalised compactness (0-1)

Eccentricity: 0.84
Elongation of fitted ellipse

Convexity: 0.902
How convex the tumor contour is

Border: Moderately defined
Border definition quality

Roughness: Moderate
Surface texture assessment

(!) IMPORTANT: This AI-generated analysis is intended for screening and research purposes only. All findings must be reviewed and validated by a qualified radiologist or neurosurgeon before any clinical decisions are made. Histopathological confirmation is required for definitive diagnosis.

1. CLINICAL INDICATION

The patient, Patient A, underwent an MRI examination to evaluate a suspected pituitary lesion. The clinical indication for this study is to assess the presence, size, and characteristics of a known or suspected pituitary tumor.

2. IMAGING TECHNIQUE

The MRI examination was performed using standard sequences, including T1-weighted and T2-weighted images, before and after contrast administration. The images were acquired in the coronal and sagittal planes to provide optimal visualization of the pituitary gland and surrounding structures.

3. FINDINGS

The MRI images reveal a mass lesion in the pituitary region, with a calculated area of 8.525 cm2 and a diameter of 3.295 cm, corresponding to an estimated volume of 4.263 cm3. The tumor appears to be moderately defined, with a heterogeneous enhancement pattern following contrast administration. The lesion is centered in the midline, with minimal mass effect, and a midline shift of 0.1 mm.

4. IMPRESSION

Based on the imaging findings, a pituitary tumor is suspected, with a confidence level of 73.68%. The tumor's characteristics, including its size and heterogeneous enhancement, are suggestive of a pituitary adenoma. However, histopathological confirmation is required to establish a definitive diagnosis.

5. SEVERITY ASSESSMENT

The tumor is classified as large based on the RANO size criteria. The risk score is 0.3/1.0, indicating a low growth risk. The tumor's irregularity is measured at 0.423, and there is no clear evidence of necrosis.

6. RECOMMENDATIONS

Given the suspected diagnosis of a pituitary tumor, further evaluation and management are recommended. Histopathological confirmation of the tumor type and grade is required, which may involve surgical intervention or biopsy. Close clinical follow-up and imaging surveillance are also recommended to monitor for potential tumor growth or changes in symptoms.